

- 7 (a) (i) R B1 [1]
- (ii) $0.5R$ B1 [1]
- (iii) $2.5R$...(*allow e.c.f. from (ii)*) B1 [1]
- (b) (i) $I_1 + I_2 = I_3$ B1 [1]
- (ii) $E_2 = I_3R + I_2R$ B1 [1]
- (iii) $E_1 - E_2 = 2I_1R - I_2R$ B1 [1]

- 8 (a) rate of decay / activity / decay (of nucleus) is not affected by external factors / environment /